module-06-moon-phases-tithi - Darshana (60-min workshop)

IKS Curriculum - senior band

Module 6 — Moon Phases and Tithi (Darśana)

Sanskrit: *Candra-pakṣa, Tithi* · **Band:** Senior (9-12) · **Duration:** 60 min · **Path:** Darśana (Workshop)

The moon's monthly cycle and the Indian lunar calendar.

> **Darśana note.** This workshop condenses Module 6's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to **Adhyayana** (the Full Course path).

Opening hook · 5 min

Show side-by-side: (a) NASA's lunar phase diagram for the current month, (b) a panchanga page for the same month. Same astronomy, two frameworks. *"How do they correspond? Where do they diverge?"*

Core idea · 15 min

Lunisolar astronomy + Indian calendar.

1. **Synodic month** (~29.530589 days, modern measurement). Indian astronomy: same value within ~0.001 days, established by Sūrya-siddhānta and refined by Bhāskara II.
2. **Tithi computation.** True tithi = elongation of Moon-Sun angle / 12°. Varies because moon's orbital speed varies (Kepler's 2nd law).
3. **Pañcāṅga five limbs.** Tithi (lunar day), vāra (weekday), nakṣatra (27 lunar mansions), yoga (Sun-Moon longitudinal sum), karaṇa (half-tithi). Each is a precise astronomical computation.
4. **Eclipse geometry.** Why solar eclipses only at new moon: Moon must be between Sun and Earth, occurring only at conjunction. Why not every new moon: lunar orbit tilted ~5.14° from ecliptic; eclipse requires near-node passage.

Demonstration · 10 min

Critical reading. Compare Sūrya-siddhānta's tithi calculation with the modern Drik Pañcāṅga (web). Both use the same underlying astronomy. Discussion: where do they differ in conventions (sidereal vs tropical reference frames)?

Source moment · 5 min

Direct verse. Sūrya-siddhānta 2.66:

> *bhāgāḥ ṣaṣṭi-guṇāḥ proktās tithayo dvi-guṇās tathā*

"Sixty divisions are stated; thus the tithis are twice that (× 60 / 12 = 30)."

The verse encodes the 12°-per-tithi rule that defines the lunar day.

Hands-on activity · 15 min

Eclipse prediction exercise. Using a published table, identify the next 3 solar and lunar eclipses visible from your region. Predict which months / tithis. Discuss: how does ancient and modern eclipse-prediction differ in precision?

Reflection + take-home · 10 min

"Indian classical astronomy achieved ~99.9% precision on synodic month calculation by ~500 CE. What does this tell us about (a) the empirical sophistication of pre-modern Indian science, (b) the limits of attempting modern claims from this evidence?"

Go deeper into Adhyayana

The 10 days you're skipping if you only do this workshop:

- **Day 01** — The moon's shape — why it changes
- **Day 02** — Pakṣa — bright and dark fortnights
- **Day 03** — 15 tithis — names and computation
- **Day 04** — Building a tithi calendar
- **Day 05** — Lunar months and festivals
- **Day 06** — Pañcāṅga — the five limbs
- **Day 07** — Eclipse geometry
- **Day 08** — Modern lunar science — phases, libration, tides
- **Day 09** — Cross-cultural calendars — Hijri, Chinese, Hebrew

- **Day 10** — Synthesis quiz + moon diary share

→ **Open the Adhyayana track for this module** to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.